

ENGR3430: Eelectronics

Mini Project 1: Hysteretic Oscillator

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Project Goals

The goal of this project is to create a USB powered circuit to flash an LED with a period of $1s \pm 10\%$. This translates to a frequency of $1Hz \pm 10\%$. The components given makes this doable by creating a hysteretic oscillator using an op-amp.

Design Decisions

Oscillator Component Value Selection

Designing this oscillation circuit required picking values for V_{ref} and the various feedback resistors and capacitors as seen in the diagram for the hysteretic oscillator circuit below.

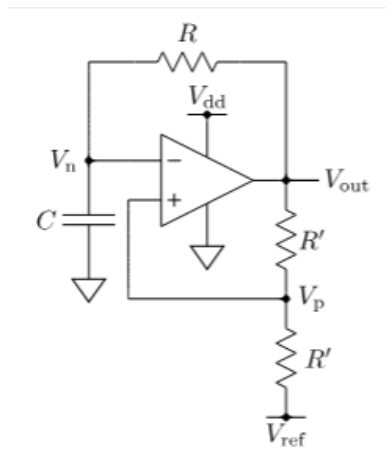


Figure 1: Overall op-amp hysteretic oscillator

Having V_{ref} set at $\frac{1}{2}V_{dd}$ allows for the upper and lower oscillator trigger thresholds to be centered in the V_{dd} range when the voltage divider resistors, R' connected to the inverting op-amp input are selected to be equal. The R' resistor values I chose were $1k\Omega$ and this was to match other resistors on the board, like the LED current limiting resistor. In this arrangement, $RC = \tau$, connected to the non-inverting op-amp input, are most important to controlling the timing of the circuit. I selected to use the $0.1\mu F$ capacitor for C because the op-amp MCP6022 IC uses a $0.1\mu F$ capacitor for its power bypass capacitance meaning the design already needs a capacitor of that size which minimized the variety of capacitor values needed. Additionally, the $0.1\mu F$ capacitor has a 5% tolerance which is less than the $1\mu F$ capacitor which has a 10% tolerance. This makes for less variation in the output frequency due to tolerances. With C fixed, I needed to solve for R and did so using the equation below in which V_{up} is the threshold in which the op-amp output goes high, V_{dn} is the threshold in which the op-amp output goes low, T_l is the period in which the oscillator outputs low.

$$V_{up} = V_{dn}e^{\frac{-T_l}{\tau}}$$

With $V_{ref} = \frac{1}{2}V_{dd}$, $V_{up} = \frac{1}{4}V_{dd}$ and $V_{dn} = \frac{3}{4}V_{dd}$. Solving for τ we get

$$\tau = \frac{-T_l}{\log\left(\frac{V_{up}}{V_{dn}}\right)}$$

Simplifying further:

$$\tau = \frac{-T_l}{\log\left(\frac{1}{3}\right)}$$

Breaking τ into RC and dividing out C solves for R :

$$R = \frac{-T_l}{\log\left(\frac{1}{3}\right) C}$$

I want the design to have a 50% duty cycle so the low time should be equal to the high time. With the design needing to have a 1Hz frequency, the high time and low time each should be 0.5s. As such $T_l = 0.5s$ and knowing that C is selected to be $0.1\mu F$, the nominal value for $R = 4.55M\Omega$. A $4.55M\Omega$ resistor is not available in the BOM but it can be closely created using a series connection of two $2M\Omega$, one $499k\Omega$, and one $49.9k\Omega$ resistors. This creates an equivalent resistance of $4.5489M\Omega$ which is less than 0.03% different than the nominal $4.55M\Omega$. Using four resistors does give more opportunity for tolerances to become an issue but it puts the equivalent resistance very close to the nominal.

The circuit runs on a single ended 3.3V supply derived from the V_{bus} of a USB using a linear regulator. As such V_{dd} for this circuit is 3.3V and since I want to have a V_{ref} equal to $\frac{1}{2}V_{dd}$, I need to divide the V_{dd} supply in two. I did this using a voltage divider and two $1k\Omega$ resistors, matching the R resistor values. I did not directly attach the V_{out} of the voltage divider to the feedback on the op-amp though. Doing this would have put the voltage divider as part of the feedback causing the 1.65V reference to vary. To mitigate this I used the second op-amp on the MCP6022 to create a buffer. I then connected the output of that op-amp for V_{ref} .

IC Application

The datasheet for the MCP1702 linear regular suggests having $1\mu F$ bypass capacitors on the input and output so these were included in the design. Additionally the MCP6022 op-amp datasheet suggests having a $0.1\mu F$ bypass capacitor within 2mm and $1\mu F$ bulk capacitor within 100mm. As such a $0.1\mu F$ cap is connected less than 2mm away from the power pin and the $1\mu F$ bypass capacitor on the output of the linear regular is within 100mm of the op-amp and acts as bulk capacitance for the op-amp.

Worse Case Analysis

I did an initial simulation in LTSpice to verify the nominal behavior. The output behavior shows a frequency of 1Hz as expected. Schematic and simulation output can be seen below:

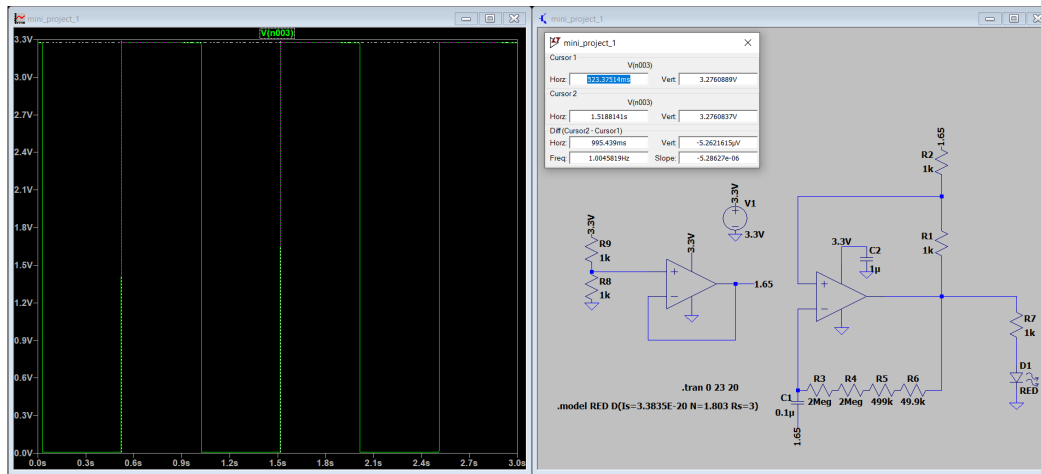


Figure 2: LTSpice schematic used for nominal behavior simulation

I also used LTSpice to do a worst-case analysis on the circuit. The simulation includes tolerance for each component and simulates the circuit against the tolerances of one another.

The LTSpice schematic can be seen below:

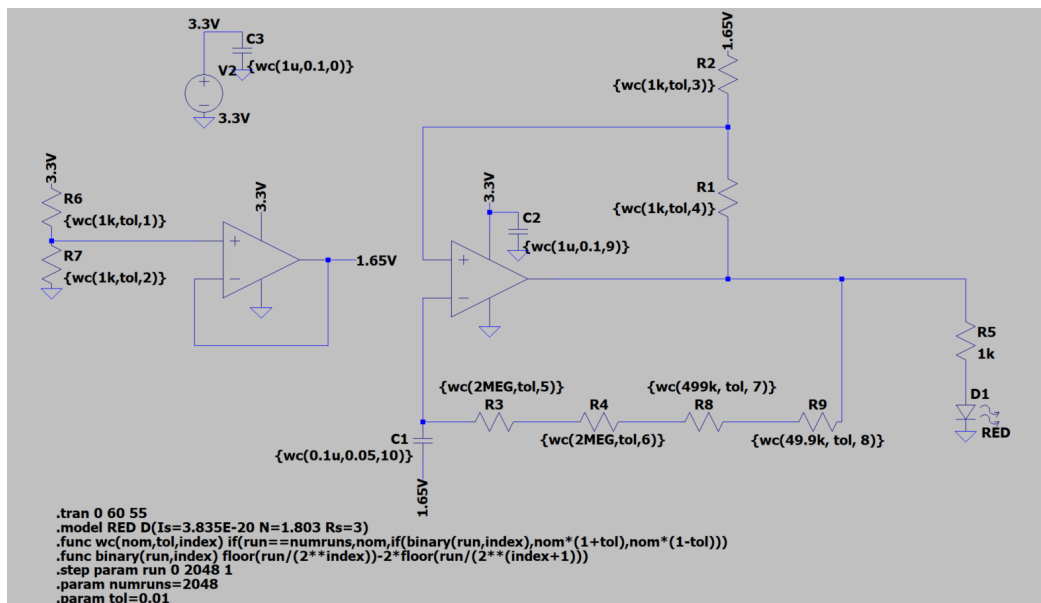


Figure 3: LTSpice schematic used for worst-case simulation

I exported the raw data for processing using Python. I created a script that uses the rising edges to calculate frequency of the signals, determines the highest and lowest frequencies of all the runs/combination, and plots all the runs on a histogram to show a distribution. My simulation uses 11 components so there were 2029 combination. Out of these, the highest frequency was 1.091Hz and the lowest was 0.935Hz both of which are within the $1\text{Hz} \pm 10\%$ requirement.

The distribution of frequencies can be seen below:

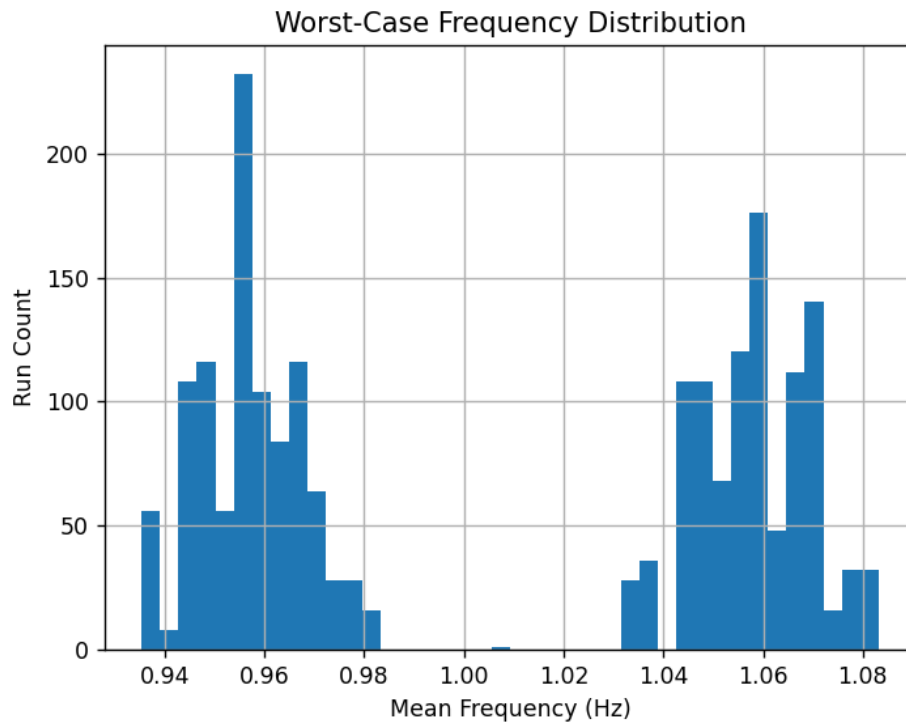


Figure 4: Distribution of frequencies in worst case simulation

Model Abstractions

One model abstraction in this simulation is not including the linear regulator. We were told that we do not need to model the tolerance of the linear regulator so I chose to generate the 3.3V power using a standard LTSpice voltage source. Since I didn't include the linear regulator directly, it also means that the bypass capacitor on the input of the linear regulator is not included. Given the nature of the bypass capacitor though, the behavior of the oscillator should

The simulation also does not include tolerance for the current limiting resistor for the LED as suggested. I did choose to include the bypass cap for the op-amp and the output of the 3.3V voltage supply but these should not influence the oscillation behavior much.

Layout Considerations

I followed the convention of having 8mil trace widths for signals and 20mil power trace widths where possible. I was able to follow this except for trace connecting to the positive rail on the op-amp. I found that 20mils was too wide and I needed to reduce it to 17mil. Although reduced in size, the current flow through that trace is still well below the max current flow for a 17mil trace so this should not cause any problems. The smaller trace width likely has a reduction in thermal capacity and bulk capacitance but not enough to cause concern for a simple circuit like this. I also used the suggested 24mil via with 12mil drill hole diameter.

I tried to focus on minimizing trace lengths and keeping routed tracing far from one another to minimize crosstalk. The frequencies in this circuit are low enough that such considerations likely are overkill but I still wanted to try to follow signal integrity best practices. I tried to keep power traces short and signals traces also short and away from unlike signals. I also tried to place vias in position to minimize loop area. My layout likely could have been made more compact but it would have been at the trade off of routing. Additionally, I wanted to consider manufacturing and assembly. Having components/traces very close to each other can make it more difficult for fabs to fabricate for a low cost and having sufficient space between can make its more viable to fabricate these on a smaller desktop style PCB mills. Additionally having done hand soldering with SMD I've learned that having little space between components increases the likelihood of solder on one pad bridging to another so the extra spacing should be helpful. I also wanted to keep sufficient space for component references. Although also potentially unnecessary, I have found them helpful when assembling boards if you don't have access to the layout.

The ERC in the layout does indicate errors due to the USB connector. The footprint includes the full outline of the port but obviously the board can't be extended that far as it would then be impossible to plug the port into a USB port. Therefore I am ignoring those errors. I edited the footprint to remove part of the footprint that extends beyond the board.